

5 Incident Response and Digital Forensics Playbook

Objective: Build and simulate the complete incident response lifecycle.

Tools

Tool / Resource	Purpose	Download / Link
TheHive	Free incident management and case tracking platform	https://thehive-project.org/
Cortex	Free automated observable analysis (pairs with TheHive)	https://github.com/TheHive-Project/Cortex
Kali Linux	Attack simulation and forensics tools	https://www.kali.org/
Volatility 3	Free memory forensics framework	https://github.com/volatilityfoundation/volatility3
Autopsy	Free digital forensics platform (disk analysis)	https://www.autopsy.com/download/
FTK Imager (free version)	Free memory and disk acquisition tool	https://www.exterro.com/ftk-imager
Wireshark	Free network traffic capture and analysis	https://www.wireshark.org/
Wazuh	Free SIEM for log collection during incident	https://wazuh.com/
Velociraptor	Free DFIR tool for endpoint investigation	https://www.rapid7.com/products/velociraptor/
GoPhish	Free phishing simulation framework	https://getgophish.com/

Incident Preparation & Setup

 Tasks	 Deliverables
• Define your IRT (Incident Response Team) roles: Incident Manager, Analyst, Forensicator, Communications Lead • Write an IR Policy: what qualifies as an incident, who is responsible, escalation path • Set up a communication plan: internal channels (email/Teams/Slack), external contacts (management, legal) • Install TheHive on Ubuntu VM for case management • Create IR playbook templates in TheHive for: Phishing, Ransomware, Data Breach • Create a pre-incident checklist: tools ready, backups verified, contacts updated, logging enabled	• IR Policy document: scope, objectives, roles, authority • Roles and responsibilities chart (table or org chart) • Communication plan document: internal contacts, escalation matrix • IR Playbook templates (in TheHive or Word format) for 2 scenarios • Pre-incident checklist document

Incident Simulation & Evidence Collection

 Tasks	 Deliverables
• Scenario 1 – Phishing: Use GoPhish to send a simulated phishing email with a fake login page to a test email • Capture the phishing email headers, landing page URL, and any credentials entered • Scenario 2 – Ransomware: Download a benign ransomware simulator (e.g., RanSim by KnowBe4) on Windows VM • Run RanSim and observe file encryption behavior (isolated VM only!) • Collect logs from Wazuh during both scenarios • Use FTK Imager to capture a memory dump (.mem file) from the Windows VM after the simulation • Capture network traffic with Wireshark during both simulations – save as .pcap files • Build an incident timeline: what happened, when, in what sequence	• Collected evidence files: email headers (.eml), memory dump (.mem), PCAP files, Wazuh logs • Incident timeline document: Timestamp Event Source Notes • Screenshots of GoPhish campaign and RanSim results • Evidence chain of custody form: who collected what, when, and how it was stored

Containment, Eradication, and Recovery

 Tasks	 Deliverables
• Containment – Phishing: Disable phishing link, block sender domain in email filter, notify all users • Containment – Ransomware: Isolate affected VM from network (disable network adapter) • Root Cause Analysis: examine Wazuh logs to trace initial access → execution → impact • Analyze memory dump with Volatility 3: <code>python3 vol.py -f memory.mem windows.pslist</code> (list processes) • Analyze PCAP with Wireshark: filter for suspicious outbound connections (known C2 IPs) • Eradication: remove malicious files, reset compromised credentials, update endpoint protection • Recovery: restore clean snapshot of VM, re-enable services, verify functionality	• Root cause analysis document: timeline, attack path, initial access vector • Volatility 3 output screenshots: process list, network connections, injected code • Containment steps document: actions taken with timestamps • Recovery steps document: what was restored and how it was verified

Post-Incident Activities

 Tasks	 Deliverables
• Conduct a lessons learned meeting: what went well, what failed, what needs improvement • Update IR playbooks based on lessons learned • Prepare executive report (1 page): what happened, business impact, actions taken, status • Prepare technical report: full incident lifecycle, evidence, analysis, remediation • Create a recommendations section: new security controls to prevent recurrence • Build PowerPoint: incident timeline, evidence walkthrough, key findings, recommendations	• Lessons learned document: improvements identified with owners and deadlines • Updated IR playbook(s) • Executive Report (1 page, non-technical language) • Technical Report (full detail) • PowerPoint Presentation: key findings and recommendations

Tips

- IMPORTANT: Run ALL simulations only in isolated VMs – never on real systems or real networks
- TheHive is very powerful for case management – it lets you track evidence, tasks, and timelines
- Volatility 3 is command-line – practice a few commands before the presentation: pslist, netscan, malfind
- FTK Imager is free and creates legally-sound forensic images – great for demonstrating proper evidence handling
- GoPhish is 100% free and open source – document your simulation scope to show it's ethical and controlled
- For ransomware simulation, use RanSim (by KnowBe4) – it simulates behavior without actually encrypting files
- Chain of custody is a key forensics concept – demonstrate you understand it even in a simulated exercise